

AEM-X100 — Wafer Loading Procedure

Equipment: AEM-X100 Wafer Handling System

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IMPORTANT: This document is entirely fictional and created for the AAS Practice Module prototype. It is not an AEM document, does not represent an actual AEM product or maintenance procedure, and must not be used for real equipment servicing.

Purpose: Define a fictional normal wafer-loading sequence and the decision points that can be used to test diagnostic routing.

1. Preconditions

- System state must be READY.
- Load port must report available.
- No active safety-interlock alarm may be present.
- Carrier and wafer identification information must be accepted by the SCU.

2. Loading Sequence

- Step 1 — Place the fictional wafer carrier at the designated load-port position.
- Step 2 — Confirm carrier presence is detected.
- Step 3 — Initiate the load sequence from the SCU.
- Step 4 — The WTM moves to the defined pickup position.
- Step 5 — WPS confirms wafer presence at the pickup position.
- Step 6 — VHM establishes the fictional handling condition.
- Step 7 — WTM transfers the wafer to the designated station.
- Step 8 — SCU confirms transfer completion and updates the wafer state.

3. If Wafer Is Not Detected

- Do not repeatedly command the transfer sequence.
- Record the displayed alarm or status message.
- Search Error Code Reference for the alarm code.
- If the alarm relates to WPS, use Sensor Troubleshooting.
- If the condition requires access to a protected area, stop and use Safety Procedure.
- If evidence remains insufficient after documented checks, use Escalation Procedure.

4. Expected Status Transitions

- READY → TRANSFER → PICKUP CONFIRMED → TRANSFER → COMPLETE.
- If a required sensor confirmation is missing, the sequence should move to a fault or paused state rather than continue indefinitely.

End of synthetic document.